

ISC Class XII Mathematics

Chapter 3

Matrices (Part A)

Matrix Algebra – Exercises 3.1 to 3.4

19 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Types Covered

- I. Order of a matrix & possible orders
- II. Constructing a matrix from a rule for a_{ij}
- III. Finding unknowns using equality of matrices ★
- IV. Counting matrices with restricted entries
- V. Direct computation – addition, subtraction, scalar multiplication
- VI. Solving $xP + yQ = R$ for unknown scalars
- VII. Simultaneous matrix equations for X and Y ★
- VIII. Checking whether a matrix product is defined
- IX. Computing a matrix product (or a single entry)
- X. Solving for an unknown via $AB = O$
- XI. Proving matrix algebra identities when $AB = BA$ ★
- XII. Using $f(A) = O$ to find higher powers of A ★
- XIII. Finding A^n by mathematical induction
- XIV. Finding an unknown matrix B from $BA = I$
- XV. Applied matrix-multiplication word problems
- XVI. Transpose properties, incl. $(AB)' = B'A'$
- XVII. Unknowns from symmetric/skew-symmetric conditions ★
- XVIII. Proving a constructed product is symmetric
- XIX. Expressing a matrix as symmetric + skew-symmetric

Compiled Notes – Formula Sheet + Type-wise Solved ISC Board Solutions

Covers Ex 3.1–3.4 only. Inverse of a matrix / solving systems of equations (Ex 3.5 onward) to follow in Part B once those pages are available.

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Index of Types

Type	Method / Pattern	★	Source(s)
1	Order of a matrix & possible orders	–	Ex 3.1 Q3
2	Constructing a matrix from a rule for a_{ij}	–	Ex 3.1 Q6(i)
3	Finding unknowns using equality of matrices	Yes	Ex 3.1 Q10; Ex 3.1 Q11 (alt.)
4	Counting matrices with restricted entries	–	Ex 3.1 Q13(i)
5	Direct computation: add/subtract/scalar mult.	–	Ex 3.2 Q4
6	Solving $xP + yQ = R$ for scalars	–	Ex 3.2 Q9
7	Simultaneous matrix equations for X, Y	Yes	Ex 3.2 Q15; Ex 3.2 Q14 (alt.)
8	Checking whether a matrix product is defined	–	Ex 3.3 Q3
9	Computing a matrix product / single entry	–	Ex 3.3 Q15(ii)
10	Solving for an unknown via $AB = O$	–	Ex 3.3 Q30(i)
11	Matrix algebra identities when $AB = BA$	Yes	Ex 3.3 Q28; Ex 3.3 Q27(i) (alt.)
12	Using $f(A) = O$ to find higher powers of A	Yes	Ex 3.3 Q31; Ex 3.3 Q24(i) (alt.)
13	Finding A^n by mathematical induction	–	Ex 3.3 Q32
14	Finding unknown matrix B from $BA = I$	–	Ex 3.3 Q33(ii)

Type	Method / Pattern	★	Source(s)
15	Applied matrix-multiplication word problems	–	Ex 3.3 Q35
16	Transpose properties, incl. $(AB)' = B'A'$	–	Ex 3.4 Q17
17	Unknowns from symmetric/skew-symmetric cond.	Yes	Ex 3.4 Q5; Ex 3.4 Q31 (alt.)
18	Proving a constructed product is symmetric	–	Ex 3.4 Q8
19	Expressing A as symmetric + skew-symmetric	–	Ex 3.4 Q30(i)

Formula Sheet

Section I – Fundamentals of Matrices

Order of a matrix

$$\text{Order} = m \times n \quad (\text{rows} \times \text{columns})$$

An $m \times n$ matrix has m rows and n columns; total entries = mn .

Equality of matrices

$$A = B \iff \text{same order, and } a_{ij} = b_{ij} \quad \forall i, j$$

Two matrices are equal only if every corresponding entry matches – not just the totals.

Section II – Addition, Subtraction & Scalar Multiplication

Addition

$$(A + B)_{ij} = a_{ij} + b_{ij}$$

Only defined when A and B have the same order; add entry by entry.

Subtraction

$$(A - B)_{ij} = a_{ij} - b_{ij}$$

Same-order requirement as addition; subtract entry by entry.

Scalar multiplication

$$(kA)_{ij} = k \cdot a_{ij}$$

Every entry of A is multiplied by the scalar k .

Section III – Multiplication of Matrices

Matrix product rule

$$(AB)_{ik} = \sum_j a_{ij} b_{jk}$$

Defined only when columns of A = rows of B; row $\times$ column, then sum.

Order of a product

$$A_{m \times n} \cdot B_{n \times p} = (AB)_{m \times p}$$

Inner dimensions must match; outer dimensions give the order of the answer.

Square of a sum (general)

$$(A + B)^2 = A^2 + AB + BA + B^2$$

True for ANY two square matrices of the same order – AB and BA cannot be merged unless they're equal.

Square of a sum (commuting)

$$(A + B)^2 = A^2 + 2AB + B^2 \quad (\text{only if } AB = BA)$$

Only valid once $AB = BA$ is proved or given – never assume it.

Square of a difference (commuting)

$$(A - B)^2 = A^2 - 2AB + B^2 \quad (\text{only if } AB = BA)$$

Same commuting condition as above.

Difference of squares (commuting)

$$(A + B)(A - B) = A^2 - B^2 \quad (\text{only if } AB = BA)$$

Same commuting condition as above.

Section IV – Transpose, Symmetric & Skew-Symmetric Matrices**Transpose properties**

$$(A')' = A, \quad (A + B)' = A' + B', \quad (kA)' = kA', \quad (AB)' = B'A'$$

Transpose of a product reverses the order of multiplication.

Symmetric matrix

$$A' = A \iff a_{ij} = a_{ji} \quad \forall i, j$$

The matrix is unchanged when reflected across the leading diagonal.

Skew-symmetric matrix

$$A' = -A \iff a_{ij} = -a_{ji} \quad \forall i, j$$

Diagonal entries must all be 0; off-diagonal entries flip sign across the diagonal.

Symmetric + skew-symmetric decomposition

$$A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$$

$\frac{1}{2}(A + A')$ is always symmetric; $\frac{1}{2}(A - A')$ is always skew-symmetric; the two always add back to A .

I. Fundamentals of Matrices

Type 1

The sum asks either for the order of a matrix directly (rows $\times$ columns), or works backwards: given the total number of entries, list every possible order the matrix could have. This is a warm-up type – no algebra, just knowing that order is a factorisation of the entry count.

Formula Used: Order = $m \times n$ (rows $\times$ columns)

Source: Exercise 3.1, Q3

Given: A matrix has 18 elements. What are the possible orders? What if it has 11 elements?

Solution:

List every pair of positive integers (m, n) such that $m \times n = 18$. This works because the order of a matrix is just how its total entries are split into rows and columns, so every factor pair of 18 is a valid order.

$$18 = 1 \times 18 = 2 \times 9 = 3 \times 6 = 6 \times 3 = 9 \times 2 = 18 \times 1$$

Each factor pair is written twice (once each way round) because a 1×18 matrix (one row) and an 18×1 matrix (one column) are genuinely different shapes, even though both hold 18 entries.

Now repeat for 11 elements. Since 11 is a prime number, it can only be factorised as 1×11 , so this checks the same idea but on a case where there's only one possible shape either way round.

$$11 = 1 \times 11 = 11 \times 1$$

Final Answer

For 18 elements: 1×18 , 18×1 , 2×9 , 9×2 , 3×6 , 6×3 .

For 11 elements: 1×11 , 11×1 .

Common Mistake: Listing only one direction of each factor pair (e.g. writing 2×9 but forgetting 9×2) – order genuinely matters, so both directions must be listed separately.

Quick Recall: The order of a matrix is just a factor pair of the total entry count – list every factor pair, and remember each pair counts twice (both ways round).

Type 2

The sum gives a rule $a_{ij} = f(i, j)$ and asks you to build the actual matrix by plugging in every valid (i, j) pair for the given order. It tests whether you correctly keep i as the row index and j as the column index while substituting.

Formula Used: $a_{ij} = f(i, j)$ substituted for every valid (i, j)

Source: Exercise 3.1, Q6(i)

Given: Construct a 2×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = 2i - j$.

Solution:

Since the matrix is 2×2 , i runs over 1, 2 (rows) and j runs over 1, 2 (columns) – so there are exactly 4 entries to compute, one for each position.

$$a_{11} = 2(1) - 1 = 1, \quad a_{12} = 2(1) - 2 = 0$$

These use $i = 1$ (first row) with $j = 1$ and $j = 2$ in turn.

$$a_{21} = 2(2) - 1 = 3, \quad a_{22} = 2(2) - 2 = 2$$

These use $i = 2$ (second row) with $j = 1$ and $j = 2$ in turn.

Now place each value at its own position (i, j) to assemble the matrix.

Final Answer

$$A = \begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$$

Common Mistake: Swapping i and j inside the formula (i.e. computing $f(j, i)$ instead of $f(i, j)$) – this only gives the wrong answer when the rule isn't symmetric in i and j , which is exactly why this rule was chosen.

Quick Recall: Just plug every row-column position into the given rule, one entry at a time – row index first, column index second.

Type 3

Two matrices are set equal to each other. Since equal matrices must match order AND have identical corresponding entries, comparing position-by-position turns the matrix equation into ordinary simultaneous equations in the unknowns, which are then solved normally.

★ **Board Favorite** – A standard high-yield pattern – comparing corresponding entries of equal matrices to set up and solve simultaneous equations is one of the most frequently repeated openers across ISC Maths papers on this chapter. I don't have confirmed exact past years for this specific sub-pattern, so I'm flagging it as a consistently high-yield type rather than citing a year.

Formula Used: $A = B \iff$ same order, and $a_{ij} = b_{ij} \forall i, j$

Source: Exercise 3.1, Q10

Given: $\begin{bmatrix} x & 3x - y \\ 2x + z & 3y - w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, find x, y, z, w .

Solution:

Compare the two matrices position by position – each pair of corresponding entries must be equal, since the matrices themselves are equal.

$$x = 3$$

This comes directly from the (1, 1) position, and immediately gives one unknown.

$$3x - y = 2 \Rightarrow 9 - y = 2 \Rightarrow y = 7$$

Substitute the already-known value of $x = 3$ into the (1, 2) equation to isolate y .

$$2x + z = 4 \Rightarrow 6 + z = 4 \Rightarrow z = -2$$

Substitute $x = 3$ into the (2, 1) equation to isolate z .

$$3y - w = 7 \Rightarrow 21 - w = 7 \Rightarrow w = 14$$

Substitute the now-known $y = 7$ into the (2, 2) equation to isolate w .

Final Answer

$$x = 3, y = 7, z = -2, w = 14$$

Common Mistake: Trying to solve equations in the order they're written rather than in the order that actually unlocks each unknown – some equations depend on a variable found in an earlier one, so solving out of sequence leads to two unknowns in one equation.

Alternate Board-Style Example

Source: Exercise 3.1, Q11

Given: $\begin{bmatrix} 2a + b & a - 2b \\ 5c - d & 4c + 3d \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$, find a, b, c, d .

Solution:

This time no single position gives an unknown directly, so pair up the equations that share the same unknowns and eliminate.

$$2a + b = 4 \dots (i), \quad a - 2b = -3 \dots (ii)$$

These come from the (1, 1) and (1, 2) positions and involve only a, b .

$$2 \times (ii) : 2a - 4b = -6 \dots (iii); \quad (i) - (iii) : 5b = 10 \Rightarrow b = 2$$

Doubling (ii) makes the a -coefficient match (i)'s, so subtracting cancels a and isolates b .

$$\text{From (i) : } 2a + 2 = 4 \Rightarrow a = 1$$

$$5c - d = 11 \dots (iv), \quad 4c + 3d = 24 \dots (v)$$

These come from the (2, 1) and (2, 2) positions and involve only c, d .

$$d = 5c - 11 \text{ (from (iv))}; \quad 4c + 3(5c - 11) = 24 \Rightarrow 19c = 57 \Rightarrow c = 3 \Rightarrow d = 4$$

Substitute (iv) into (v) to reduce to a single unknown c , then back-substitute for d .

Final Answer

$$a = 1, \quad b = 2, \quad c = 3, \quad d = 4$$

Quick Recall: Equal matrices means same size AND every position matches – line up the entries and solve exactly like ordinary simultaneous equations.

Type 4

The sum asks how many different matrices of a given order can be built if each entry is restricted to a fixed small set of values (e.g. only 0 or 1, or only 1, 2, or 3). Every entry is an independent choice, so the total count multiplies out.

Formula Used: Total matrices = (number of choices per entry)^(total number of entries)

Source: Exercise 3.1, Q13(i)

Given: Write the number of all possible matrices of order 2×3 with each entry 1 or 2.

Solution:

First count the total number of entries in the matrix, since each one is an independent slot to fill.

$$\text{Total entries} = 2 \times 3 = 6$$

Each of these 6 entries can independently be filled in 2 ways (either 1 or 2), so the total number of distinct matrices is 2 multiplied by itself once per entry.

$$\text{Total matrices} = 2^6 = 64$$

Final Answer

64

Common Mistake: Flipping the base and exponent – writing (number of entries)^(number of choices) instead of (number of choices)^(number of entries). It's the number of independent slots that goes in the exponent.

Quick Recall: Each entry is an independent choice – raise the number of choices to the power of the total number of entries.

II. Addition, Subtraction & Scalar Multiplication

Type 5

Straightforward computation: scale and/or combine given matrices of the same order entry by entry. No unknowns to solve for – just careful, position-by-position arithmetic.

Formula Used: $(kA)_{ij} = k \cdot a_{ij}$, $(A \pm B)_{ij} = a_{ij} \pm b_{ij}$

Source: Exercise 3.2, Q4

Given: $A = \begin{bmatrix} 1 & 5 \\ -2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 4 & -7 \end{bmatrix}$, find $2A - 3B$.

Solution:

Scale each matrix by its own scalar first – this must happen before combining, since $2A - 3B$ is not the same as $2(A - B)$.

$$2A = \begin{bmatrix} 2 & 10 \\ -4 & 6 \end{bmatrix}, \quad 3B = \begin{bmatrix} 0 & -3 \\ 12 & -21 \end{bmatrix}$$

Now subtract the two scaled matrices entry by entry, since both are the same 2×2 order and so are compatible for subtraction.

$$2A - 3B = \begin{bmatrix} 2 - 0 & 10 - (-3) \\ -4 - 12 & 6 - (-21) \end{bmatrix} = \begin{bmatrix} 2 & 13 \\ -16 & 27 \end{bmatrix}$$

Final Answer

$$\begin{bmatrix} 2 & 13 \\ -16 & 27 \end{bmatrix}$$

Common Mistake: Subtracting the unscaled matrices first and scaling afterwards, or scaling only one of the two matrices – both scalars must be applied before any combining happens.

Quick Recall: Scale first, combine second – multiply every entry by its scalar before adding or subtracting matrices.

Type 6

An equation of the form $xP + yQ = R$ is given, where P, Q, R are column (or row) vectors and x, y are unknown scalars. Expanding the scalar multiplications and com-

paring corresponding entries turns this into two simultaneous linear equations in x and y .

Formula Used: $xP + yQ = R \Rightarrow$ compare each row to get simultaneous equations in x, y

Source: Exercise 3.2, Q9

Given: $x \begin{bmatrix} 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$, find x and y .

Solution:

Expand the scalar multiplications on the left, then compare the top and bottom entries separately – since these are column vectors, each row gives its own equation.

$$2x - y = 10 \dots (i) \quad (\text{top row})$$

$$3x + y = 5 \dots (ii) \quad (\text{bottom row})$$

Add (i) and (ii) directly – the y terms have opposite signs, so they cancel and leave a single equation in x .

$$5x = 15 \Rightarrow x = 3$$

Substitute $x = 3$ back into (ii) to find y .

$$3(3) + y = 5 \Rightarrow y = -4$$

Final Answer

$$x = 3, y = -4$$

Common Mistake: Mixing up which row gives which equation, or dropping a sign when reading off the equation from a column entry that already had a minus sign built into the vector (like the -1 here).

Quick Recall: Expand the scalars against each row of the column vectors – that turns it into two ordinary simultaneous equations in x and y .

Type 7

Two equations link two unknown matrices X and Y (e.g. $2X - Y = \dots$ and $X + 2Y = \dots$). Exactly like solving simultaneous linear equations, multiply one or both equations by suitable scalars so that X (or Y) cancels when the equations are added or subtracted, then back-substitute.

★ **Board Favorite** – A standard high-yield pattern – “find matrices X and Y satisfying two simultaneous matrix equations” is a recurring type across ISC papers on this chapter; I don’t have confirmed exact years for this specific pattern, so flagging it as consistently high-yield rather than citing a year.

Formula Used: Eliminate one unknown matrix by scaling and adding/subtracting the two equations, just as with simultaneous linear equations

Source: Exercise 3.2, Q15

Given: $2X - Y = \begin{bmatrix} 6 & -6 & 0 \\ -4 & 2 & 1 \end{bmatrix} \dots (i)$, $X + 2Y = \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix} \dots (ii)$. Find X and Y.

Solution:

To eliminate Y, make its coefficient in (i) match the one in (ii): multiply (i) by 2 so that its Y-coefficient becomes -2 , matching (ii)'s $+2$ with an opposite sign.

$$2 \times (i) : 4X - 2Y = \begin{bmatrix} 12 & -12 & 0 \\ -8 & 4 & 2 \end{bmatrix} \dots (iii)$$

Now add (iii) and (ii) – the Y terms cancel ($-2Y + 2Y = 0$), leaving an equation purely in X.

$$5X = \begin{bmatrix} 15 & -10 & 5 \\ -10 & 5 & -5 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 1 & -1 \end{bmatrix}$$

Divide every entry by 5 to isolate X.

Substitute this X back into (ii) to solve for Y.

$$2Y = \begin{bmatrix} 3 & 2 & 5 \\ -2 & 1 & -7 \end{bmatrix} - X = \begin{bmatrix} 0 & 4 & 4 \\ 0 & 0 & -6 \end{bmatrix} \Rightarrow Y = \begin{bmatrix} 0 & 2 & 2 \\ 0 & 0 & -3 \end{bmatrix}$$

Final Answer

$$X = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 1 & -1 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & 2 & 2 \\ 0 & 0 & -3 \end{bmatrix}$$

Common Mistake: Choosing a multiplier that doesn't actually make the coefficients match before adding/subtracting, or making an entry-wise arithmetic slip while subtracting the matrix X back out of equation (ii).

Alternate Board-Style Example

Source: Exercise 3.2, Q14

Given: $X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$, $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$. Find X and Y.

Solution:

Here the coefficients of X already match (both are $+X$), so simply adding the two equations directly cancels Y.

$$2X = \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$$

Similarly, subtracting the second equation from the first cancels X and isolates Y.

$$2Y = \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} \Rightarrow Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

Final Answer

$$X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}, \quad Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

Quick Recall: Treat the two matrix equations exactly like two simultaneous linear equations – scale to match coefficients, add or subtract to cancel one matrix, then back-substitute.

III. Multiplication of Matrices

Type 8

Before multiplying two matrices, check whether the product is even defined: the number of columns in the first matrix must equal the number of rows in the second. This type only asks for that check (and the resulting order), not the actual product.

Formula Used: $A_{m \times n} \cdot B_{n \times p}$ is defined $\iff$ columns of A = rows of B

Source: Exercise 3.3, Q3

Given: A is a matrix of order 3×3 and B is a matrix of order 2×3 . Can you find AB?

Solution:

Compare the number of columns of A to the number of rows of B, since that's the only requirement for a product to be defined.

$$\text{Columns of A} = 3, \quad \text{Rows of B} = 2$$

Since $3 \neq 2$, the inner dimensions don't match, so the product AB cannot be formed at all – there's no order to compute because the multiplication itself is undefined.

Final Answer

No, AB cannot be found (columns of A $\neq$ rows of B).

Common Mistake: Assuming that because both matrices are "compatible looking" (same number of rows, or same total entries), the product must be defined – the ONLY rule that matters is columns-of-first = rows-of-second.

Quick Recall: Before multiplying, just check: does the number of columns in the first matrix equal the number of rows in the second? If not, stop – the product doesn't exist.

Type 9

A genuine matrix product has to be computed (possibly combined with addition/subtraction afterwards). Each entry of the answer is built using the row-by-column rule: multiply corresponding entries of a row of the first matrix and a column of the second, then add them up.

Formula Used: $(AB)_{ik} = \sum_j a_{ij} b_{jk}$

Source: Exercise 3.3, Q15(ii)

Given: Write as a single matrix: $\begin{bmatrix} 3 & 2 & 5 \\ 7 & -4 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & -1 \\ 3 & 5 \end{bmatrix} - \begin{bmatrix} 7 & -8 \\ 5 & 9 \end{bmatrix}$

Solution:

First compute the product of the two matrices, since multiplication is done before the subtraction. The first matrix is 2×3 and the second is 3×2 , so the product is 2×2 .

For the (1, 1) entry, take row 1 of the first matrix and column 1 of the second, multiplying corresponding entries and summing.

$$(1, 1) : 3(2) + 2(2) + 5(3) = 6 + 4 + 15 = 25$$

Repeat the same row-by-column rule for the remaining three positions.

$$(1, 2) : 3(2) + 2(-1) + 5(5) = 6 - 2 + 25 = 29$$

$$(2, 1) : 7(2) + (-4)(2) + 0(3) = 14 - 8 + 0 = 6$$

$$(2, 2) : 7(2) + (-4)(-1) + 0(5) = 14 + 4 + 0 = 18$$

So the product matrix is:

$$\begin{bmatrix} 25 & 29 \\ 6 & 18 \end{bmatrix}$$

Now subtract the given matrix entry by entry, since both matrices are now the same 2×2 order.

$$\begin{bmatrix} 25 & 29 \\ 6 & 18 \end{bmatrix} - \begin{bmatrix} 7 & -8 \\ 5 & 9 \end{bmatrix} = \begin{bmatrix} 18 & 37 \\ 1 & 9 \end{bmatrix}$$

Final Answer

$$\begin{bmatrix} 18 & 37 \\ 1 & 9 \end{bmatrix}$$

Common Mistake: Multiplying entry-by-entry as if it were addition (pairing up same-position entries) instead of using the row-by-column dot-product rule – matrix multiplication is never "position-wise" like addition is.

Quick Recall: Every entry of a product is a row of the first matrix "dotted" with a column of the second – multiply and add, one entry at a time.

Type 10

A product of matrices (possibly a row vector, a square matrix, and a column vector chained together) is set equal to the zero matrix, and an unknown inside one of the vectors has to be found. Multiply the matrices out step by step (always multiplying adjacent, compatible matrices first), then solve the resulting equation in the unknown.

Formula Used: $(AB)_{ik} = \sum_j a_{ij} b_{jk}$, applied repeatedly, then set the final entry = 0

Source: Exercise 3.3, Q30(i)

Given: Find x , if $\begin{bmatrix} 1 & x & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = O$

Solution:

There are three matrices chained together, so multiply adjacent ones first. Here it's cleanest to multiply the 3×3 matrix with the column vector first, since both are fully numeric (no unknown mixed into a row/column relationship yet beyond x itself).

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ x \end{bmatrix} = \begin{bmatrix} 1(1) + 3(2) + 2x \\ 2(1) + 5(2) + 1x \\ 15(1) + 3(2) + 2x \end{bmatrix} = \begin{bmatrix} 7 + 2x \\ 12 + x \\ 21 + 2x \end{bmatrix}$$

Each row of the square matrix is dotted with the column vector, using the standard row-by-column rule.

Now multiply the row vector $\begin{bmatrix} 1 & x & 1 \end{bmatrix}$ with this resulting column vector – this collapses everything to a single number, which must equal 0.

$$1(7 + 2x) + x(12 + x) + 1(21 + 2x) = 0$$

Expand and collect like terms: the x -terms from all three pieces combine, and the constants combine separately.

$$7 + 2x + 12x + x^2 + 21 + 2x = 0 \Rightarrow x^2 + 16x + 28 = 0$$

This is now an ordinary quadratic in x , solved using the quadratic formula.

$$x = \frac{-16 \pm \sqrt{16^2 - 4(1)(28)}}{2} = \frac{-16 \pm \sqrt{256 - 112}}{2} = \frac{-16 \pm 12}{2}$$

$$x = -2 \quad \text{or} \quad x = -14$$

Final Answer

$$x = -2 \text{ or } x = -14$$

Common Mistake: Trying to multiply the row vector and column vector first (skipping the middle matrix), which isn't a valid multiplication order here – adjacent, compatible matrices must be multiplied first, working through the chain in sequence.

Quick Recall: Multiply the chain of matrices two at a time (always adjacent, compatible pairs), collapse down to a single equation, then solve for the unknown like an ordinary equation.

Type 11

Two square matrices A and B of the same order are given, with the extra condition $AB = BA$ (they "commute"). The sum asks to expand and prove an algebraic identity such as $(A + B)^2 = A^2 + 2AB + B^2$ – something that looks like ordinary algebra but is only valid here because AB and BA have been shown (or given) to be equal.

★ **Board Favorite** – A standard high-yield pattern – proving matrix-algebra identities under the condition $AB = BA$ (and testing whether students wrongly apply them when $AB \neq BA$) is one of the most frequently repeated proof-style questions on this chapter across ISC papers. I don't have confirmed exact years for this specific pattern, so I'm calling it consistently high-yield rather than citing a year.

Formula Used: $(A + B)^2 = A^2 + AB + BA + B^2$ (always true, general expansion)

Formula Used: $(A + B)^2 = A^2 + 2AB + B^2$ (only once $AB = BA$ is used)

Source: Exercise 3.3, Q28(ii)

Given: If A and B are square matrices of the same order such that $AB = BA$, prove that $(A + B)^2 = A^2 + 2AB + B^2$.

Solution:

Start by writing $(A + B)^2$ as a product, since "squaring" a matrix means multiplying it by itself.

$$(A + B)^2 = (A + B)(A + B)$$

Expand using the distributive law of matrix multiplication over addition – this step is always valid, regardless of whether A and B commute.

$$= A(A + B) + B(A + B) = AA + AB + BA + BB$$

Now, and only now, use the given condition $AB = BA$ to replace BA with AB – this is the one step that is NOT automatically true for all matrices.

$$= A^2 + AB + AB + B^2 = A^2 + 2AB + B^2$$

Final Answer

$(A + B)^2 = A^2 + 2AB + B^2$ – proved, using $AB = BA$.

Common Mistake: Treating this identity as always true for any two matrices – without $AB = BA$ explicitly given (or proved), the correct expansion stays as $A^2 + AB + BA + B^2$, and merging $AB + BA$ into $2AB$ without justification is the single most common slip in this type.

Alternate Board-Style Example

Source: Exercise 3.3, Q27(i)

Given: Let A and B be square matrices of the same order. Does $(A + B)^2 = A^2 + 2AB + B^2$ hold? If not, why?

Solution:

Expand $(A + B)^2$ the same way as before, purely by the distributive law – this part never depends on whether A and B commute.

$$(A + B)^2 = (A + B)(A + B) = A^2 + AB + BA + B^2$$

Here, no condition like $AB = BA$ has been given, so AB and BA cannot be assumed equal and therefore cannot be merged into $2AB$.

Final Answer

No, the identity does not hold in general. The correct expansion is $A^2 + AB + BA + B^2$, since matrix multiplication is not commutative unless stated otherwise.

Quick Recall: Expand $(A + B)^2$ by distributing like normal algebra, but never merge AB and BA into $2AB$ unless you're told (or have proved) that $AB = BA$.

Type 12

A square matrix A is given along with a quadratic $f(x) = x^2 - \lambda x + \mu$, and the sum first asks to verify $f(A) = O$. The real point of the type is what comes after: using $f(A) = O$ (rearranged to $A^2 = \lambda A - \mu I$) to find higher powers of A like A^3 or A^5 WITHOUT multiplying the matrix out repeatedly by brute force.

★ **Board Favorite** – A standard high-yield pattern – using a matrix's own polynomial identity ($f(A) = O$) to efficiently compute higher powers is a classic, frequently repeated ISC pattern for 2×2 matrices. I don't have confirmed exact years for this exact sub-type, so flagging it as consistently high-yield rather than citing a year.

Formula Used: $f(A) = A^2 - \lambda A + \mu I = O \Rightarrow A^2 = \lambda A - \mu I$

Source: Exercise 3.3, Q31

Given: $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$, $f(x) = x^2 - 4x + 7$. Show that $f(A) = O$. Use this to find A^3 and A^5 .

Solution:

First compute A^2 directly, since it's needed to test the polynomial condition.

$$A^2 = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 4 - 3 & 6 + 6 \\ -2 - 2 & -3 + 4 \end{bmatrix} = \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix}$$

Now substitute into $f(A) = A^2 - 4A + 7I$ and check whether it collapses to the zero matrix.

$$f(A) = \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} - \begin{bmatrix} 8 & 12 \\ -4 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

This confirms A satisfies its own quadratic, which is exactly the tool needed to simplify higher powers.

Rearranging $f(A) = O$ gives $A^2 = 4A - 7I$ – this lets any higher power of A be rewritten purely in terms of A and I , avoiding repeated heavy multiplication.

$$A^3 = A^2 \cdot A = (4A - 7I)A = 4A^2 - 7A$$

Substitute $A^2 = 4A - 7I$ again, since A^3 still contains an A^2 term.

$$= 4(4A - 7I) - 7A = 16A - 28I - 7A = 9A - 28I$$

Now substitute the actual matrix values of A and I .

$$A^3 = 9 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 28 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 18 - 28 & 27 \\ -9 & 18 - 28 \end{bmatrix} = \begin{bmatrix} -10 & 27 \\ -9 & -10 \end{bmatrix}$$

For A^5 , write it as $A^3 \cdot A^2$ and substitute both reduced forms so the whole product stays in terms of A and I .

$$A^5 = A^3 \cdot A^2 = (9A - 28I)(4A - 7I) = 36A^2 - 63A - 112A + 196I = 36A^2 - 175A + 196I$$

Substitute $A^2 = 4A - 7I$ one more time to fully reduce.

$$= 36(4A - 7I) - 175A + 196I = 144A - 252I - 175A + 196I = -31A - 56I$$

$$A^5 = -31 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 56 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -62 - 56 & -93 \\ 31 & -62 - 56 \end{bmatrix} = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}$$

Final Answer

$$A^3 = \begin{bmatrix} -10 & 27 \\ -9 & -10 \end{bmatrix}, \quad A^5 = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix}$$

Common Mistake: Ignoring the reduction relation entirely and instead multiplying A by itself repeatedly (computing A^2, A^3, A^4, A^5 one at a time by brute force) – slow and much more error-prone than substituting $A^2 = 4A - 7I$ back in at each stage.

Alternate Board-Style Example

Source: Exercise 3.3, Q24(i)

Given: $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, show that $f(A) = O$, where $f(x) = x^2 - 2x - 3$.

Solution:

Compute A^2 first.

$$A^2 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+4 & 2+2 \\ 2+2 & 4+1 \end{bmatrix} = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$$

Substitute into $f(A) = A^2 - 2A - 3I$.

$$f(A) = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} - \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

Final Answer

$f(A) = O$ – verified.

Quick Recall: If a matrix satisfies its own polynomial equation, solve that equation for A^2 and keep substituting it back in – never multiply matrices out by brute force to get higher powers.

Type 13

A formula for A^n is given (in terms of n), and it must be proved for all positive integers using the principle of mathematical induction: verify the base case $n = 1$, assume the formula holds for $n = k$, then show it must also hold for $n = k + 1$.

Formula Used: Principle of Mathematical Induction: verify $P(1)$, assume $P(k)$, prove $P(k + 1)$

Source: Exercise 3.3, Q32

Given: $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, prove that $A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$ for all positive integers n .

Solution:

Base case ($n = 1$): check the formula against the matrix itself.

$$A^1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2^0 & 2^0 \\ 2^0 & 2^0 \end{bmatrix} \quad (\text{since } 2^0 = 1)$$

The formula matches for $n = 1$, so the base case holds.

Inductive hypothesis: assume the formula is true for some $n = k$, i.e.

$$A^k = \begin{bmatrix} 2^{k-1} & 2^{k-1} \\ 2^{k-1} & 2^{k-1} \end{bmatrix}$$

This assumption is what will be used (not proved again) to build the $n = k + 1$ case.

Inductive step: show the formula also holds for $n = k + 1$, by writing $A^{k+1} = A^k \cdot A$ and multiplying out using the assumed form of A^k .

$$A^{k+1} = \begin{bmatrix} 2^{k-1} & 2^{k-1} \\ 2^{k-1} & 2^{k-1} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2^{k-1} & 2 \cdot 2^{k-1} \\ 2 \cdot 2^{k-1} & 2 \cdot 2^{k-1} \end{bmatrix} = \begin{bmatrix} 2^k & 2^k \\ 2^k & 2^k \end{bmatrix}$$

Every entry of the product is the sum of two equal terms $2^{k-1} + 2^{k-1}$, which combines to $2 \cdot 2^{k-1} = 2^k$ – exactly the form the formula predicts for $n = k + 1$ (since $(k + 1) - 1 = k$).

Since the formula holds for $n = 1$, and its truth for $n = k$ forces its truth for $n = k + 1$, by the principle of mathematical induction it holds for all positive integers n .

Final Answer

$A^n = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}$ for all positive integers n – proved by induction.

Common Mistake: Skipping the explicit base-case check and jumping straight to the inductive step, or an algebra slip while combining exponents – writing $2^{k-1} + 2^{k-1}$ incorrectly as 2^{2k-2} instead of correctly as $2 \cdot 2^{k-1} = 2^k$.

Quick Recall: Check it works for $n = 1$, assume it works for $n = k$, then multiply by A one more time to show it must also work for $n = k + 1$.

Type 14

A matrix A is given, and an unknown matrix B must be found so that a stated product condition holds (e.g. $BA = I$). Since B 's entries are unknown, set $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, multiply out symbolically, and compare with the target matrix to get equations in a, b, c, d .

Formula Used: $(BA)_{ik} = \sum_j b_{ij}a_{jk}$, then compare entry-by-entry with the target matrix

Source: Exercise 3.3, Q33(ii)

Given: $A = \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix}$, find matrix B such that $BA = I$.

Solution:

Let $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, since B 's entries are exactly what needs to be found. Multiply B and A symbolically using the row-by-column rule.

$$BA = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 3 & -4 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 3a - b & -4a + 2b \\ 3c - d & -4c + 2d \end{bmatrix}$$

Set this equal to $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and compare entry by entry – this gives four equations, which split naturally into two independent pairs: one in (a, b) , one in (c, d) .

$$3a - b = 1 \quad \dots (i), \quad -4a + 2b = 0 \quad \dots (ii)$$

$$3c - d = 0 \quad \dots (iii), \quad -4c + 2d = 1 \quad \dots (iv)$$

From (ii): $b = 2a$. Substituting into (i) isolates a , then b follows immediately.

$$3a - 2a = 1 \Rightarrow a = 1 \Rightarrow b = 2$$

From (iii): $d = 3c$. Substituting into (iv) isolates c , then d follows immediately.

$$-4c + 2(3c) = 1 \Rightarrow 2c = 1 \Rightarrow c = \frac{1}{2} \Rightarrow d = \frac{3}{2}$$

Final Answer

$$B = \begin{bmatrix} 1 & 2 \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix}$$

Common Mistake: Assuming $BA = I$ means B can be written down directly as "the inverse of A " using a shortcut formula – at this stage of the chapter, B must be found by setting up and solving the linear equations directly, not by any inverse-matrix shortcut.

Quick Recall: Let B have unknown entries a, b, c, d , multiply out BA symbolically, then compare with the target matrix entry-by-entry to solve for each unknown.

Type 15

A real-world scenario (buying items at different prices, exchanging goods, etc.) is described, and the numbers must be arranged into two compatible matrices so that their product gives a meaningful total (e.g. total bill). This tests setting up the right matrices in the right order, not just multiplying.

Formula Used: Represent quantities as one matrix and rates/prices as another, ordered so their product is defined and meaningful

Source: Exercise 3.3, Q35

Given: Ram buys 2 dozen notebooks, 1 dozen pens, 4 pencils; Mohan buys 1 dozen notebooks, 6 pens, 8 pencils; Ankit buys 6 notebooks, 4 pens, 6 pencils. A notebook costs Rs. 45, a pen Rs. 13.50, a pencil Rs. 4.50. Find the order of the purchase matrix A , the price matrix B , and the product AB (or the total bill).

Solution:

First convert every quantity into the same unit (individual items), since "dozens" can't be mixed directly with single-item prices.

$$2 \text{ dozen} = 24, \quad 1 \text{ dozen} = 12$$

Now build the purchase matrix A , with one row per student and one column per item type, so that each row can later be matched against a column of prices.

$$A = \begin{bmatrix} 24 & 12 & 4 \\ 12 & 6 & 8 \\ 6 & 4 & 6 \end{bmatrix} \quad (\text{rows: Ram, Mohan, Ankit; columns: notebooks, pens, pencils})$$

Build the price matrix B as a column, since prices are per single item and need to line up with A's columns for multiplication.

$$B = \begin{bmatrix} 45 \\ 13.5 \\ 4.5 \end{bmatrix}$$

Since A is 3×3 and B is 3×1 , the inner dimensions ($3 = 3$) match, so AB is defined and has order 3×1 – one total per student.

$$\text{Order of B} = 3 \times 1, \quad \text{Order of A} = 3 \times 3, \quad \text{Order of AB} = 3 \times 1$$

Now compute AB itself to get each student's individual bill, using the row-by-column rule.

$$\text{Ram: } 24(45) + 12(13.5) + 4(4.5) = 1080 + 162 + 18 = 1260$$

$$\text{Mohan: } 12(45) + 6(13.5) + 8(4.5) = 540 + 81 + 36 = 657$$

$$\text{Ankit: } 6(45) + 4(13.5) + 6(4.5) = 270 + 54 + 27 = 351$$

Add the three totals to get the combined bill for all three students.

$$\text{Total bill} = 1260 + 657 + 351 = 2268$$

Final Answer

Order of B = 3×1 ; order of A = 3×3 ; order of AB = 3×1 ; total bill by all three students = Rs. 2268

Common Mistake: Multiplying in the wrong order (computing BA instead of AB) – here BA isn't even defined (1×3 columns don't match A's 3×3 rows the right way), so the purchase matrix must always come before the price matrix in this setup.

Quick Recall: Put quantities in one matrix and prices in a column next to it, in that order, so the product directly gives each person's total spend.

IV. Transpose, Symmetric & Skew-Symmetric Matrices

Type 16

Given two matrices A and B (often a column and a row vector, or two general matrices), the sum asks to verify a transpose property – most commonly $(AB)' = B'A'$: compute both sides independently and confirm they match. The key idea is that transposing a product reverses the multiplication order.

Formula Used: $(AB)' = B'A'$ (transpose of a product reverses the order)

Source: Exercise 3.4, Q17

Given: $A = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix}$, $B = [-1 \ 2 \ 1]$, verify that $(AB)' = B'A'$.

Solution:

First compute AB directly. Since A is 3×1 and B is 1×3 , the product is a 3×3 matrix.

$$AB = \begin{bmatrix} 1 \\ -4 \\ 3 \end{bmatrix} [-1 \ 2 \ 1] = \begin{bmatrix} -1 & 2 & 1 \\ 4 & -8 & -4 \\ -3 & 6 & 3 \end{bmatrix}$$

Each row of AB is just A's corresponding entry multiplying the whole row B.

Now transpose this result by swapping rows and columns.

$$(AB)' = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$$

Next compute the right-hand side, $B'A'$ – note the order is reversed compared to AB.

$$B' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad A' = [1 \ -4 \ 3]$$

$$B'A' = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} [1 \ -4 \ 3] = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix}$$

Comparing the two results shows they are identical, confirming the property.

Final Answer

$$(AB)' = B'A' = \begin{bmatrix} -1 & 4 & -3 \\ 2 & -8 & 6 \\ 1 & -4 & 3 \end{bmatrix} \text{ -- verified.}$$

Common Mistake: Computing $A'B'$ (multiplying the transposes in the ORIGINAL order) instead of $B'A'$ – the whole point of this identity is that the order reverses, and this is the single most common slip.

Quick Recall: Transposing a product always reverses the order of multiplication – $(AB)'$ becomes $B'A'$, not $A'B'$.

Type 17

A matrix contains one or more unknowns, and it's stated to be symmetric or skew-symmetric. Using the definition – symmetric means mirror-image entries are equal ($a_{ij} = a_{ji}$), skew-symmetric means they're negatives of each other ($a_{ij} = -a_{ji}$, diagonal entries all 0) – gives direct equations for the unknowns.

★ **Board Favorite** – A confirmed board favorite – this exact pattern (find an unknown from a symmetric or skew-symmetric condition) has appeared on ISC papers in 2017, 2018, 2019, 2023, and 2025, per the year-tags on the source questions themselves.

Formula Used: Symmetric: $a_{ij} = a_{ji} \forall i, j$ Skew-symmetric: $a_{ij} = -a_{ji} \forall i, j$

Source: Exercise 3.4, Q5 (ISC 2023)

Given: For what value of k is the matrix $\begin{bmatrix} 0 & k \\ -6 & 0 \end{bmatrix}$ skew-symmetric?

Solution:

For a matrix to be skew-symmetric, every off-diagonal pair of mirror-image entries must be negatives of each other – that's the direct definition being applied here.

$$a_{12} = -a_{21}$$

Here $a_{12} = k$ and $a_{21} = -6$, so substitute directly into the condition.

$$k = -(-6) = 6$$

(The diagonal entries are already both 0, which is automatically consistent with skew-symmetry.)

Final Answer

$$k = 6$$

Common Mistake: Setting $a_{12} = a_{21}$ (the symmetric condition) instead of $a_{12} = -a_{21}$ (the skew-symmetric condition) – confusing the sign in the two definitions is the single most common error in this type.

Alternate Board-Style Example

Source: Exercise 3.4, Q31 (ISC 2025)

Given: For what value of x is $A = \begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ x & -3 & 0 \end{bmatrix}$ a skew-symmetric matrix?

Solution:

Check every off-diagonal mirror pair against the skew-symmetric condition $a_{ij} = -a_{ji}$. The pair (1, 2) and (2, 1) gives 1 and -1 , which already satisfies $1 = -(-1)$; the pair (2, 3) and (3, 2) gives 3 and -3 , which already satisfies $3 = -(-3)$. Only the pair involving x remains to be checked.

$$a_{31} = -a_{13} \Rightarrow x = -(-2) = 2$$

Final Answer

$$x = 2$$

Quick Recall: Symmetric = flip across the diagonal and nothing changes; skew-symmetric = flip and everything flips sign (with zeros forced on the diagonal) – match entries to the right definition.

Type 18

A general square matrix A (no specific numbers) is given, and a constructed product like $A'A$ must be proved symmetric. This uses only the transpose rules abstractly – show that the transpose of the constructed matrix equals the matrix itself.

Formula Used: $(AB)' = B'A'$ and $(A')' = A$

Source: Exercise 3.4, Q8

Given: If A is a square matrix, prove that $A'A$ is symmetric.

Solution:

Let $M = A'A$. To prove M is symmetric, it must be shown that $M' = M$.

$$M' = (A'A)'$$

View $A'A$ as a product of two factors, $X = A'$ and $Y = A$, and apply the reversal rule $(XY)' = Y'X'$ – transposing a product swaps both the order AND transposes each factor.

$$M' = (A'A)' = Y'X' = A' \cdot (A')'$$

Since transposing twice returns the original matrix, $(A')' = A$, so this simplifies directly.

$$M' = A' \cdot A = A'A = M$$

Since $M' = M$, the matrix $M = A'A$ satisfies the definition of a symmetric matrix.

Final Answer

$A'A$ is symmetric – proved.

Common Mistake: Misapplying the reversal rule and writing $(A'A)'$ as AA' or $A'A'$ by reversing "on sight" instead of carefully labelling which factor is X and which is Y before substituting into $(XY)' = Y'X'$.

Quick Recall: To prove a matrix is symmetric, transpose it using $(PQ)' = Q'P'$ and show you land back on the same matrix you started with.

Type 19

Any square matrix A can be split into a symmetric part and a skew-symmetric part that add back to A. The sum asks to actually compute this decomposition for a given matrix, using $\frac{1}{2}(A + A')$ for the symmetric part and $\frac{1}{2}(A - A')$ for the skew-symmetric part.

Formula Used: $A = \frac{1}{2}(A + A') + \frac{1}{2}(A - A')$

Source: Exercise 3.4, Q30(i)

Given: Express $A = \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

Solution:

First find the transpose of A, since both parts of the decomposition need it.

$$A' = \begin{bmatrix} 3 & 4 & 0 \\ 2 & 1 & 6 \\ 5 & 3 & 7 \end{bmatrix}$$

Compute $A + A'$, then halve it to get the symmetric part P . This half-sum is always symmetric because transposing it just swaps the two matrices being added, which doesn't change the sum.

$$A + A' = \begin{bmatrix} 6 & 6 & 5 \\ 6 & 2 & 9 \\ 5 & 9 & 14 \end{bmatrix} \Rightarrow P = \frac{1}{2}(A + A') = \begin{bmatrix} 3 & 3 & 2.5 \\ 3 & 1 & 4.5 \\ 2.5 & 4.5 & 7 \end{bmatrix}$$

Compute $A - A'$, then halve it to get the skew-symmetric part Q . This half-difference is always skew-symmetric because transposing it swaps the order of subtraction, flipping

the sign of the whole matrix.

$$A - A' = \begin{bmatrix} 0 & -2 & 5 \\ 2 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix} \Rightarrow Q = \frac{1}{2}(A - A') = \begin{bmatrix} 0 & -1 & 2.5 \\ 1 & 0 & -1.5 \\ -2.5 & 1.5 & 0 \end{bmatrix}$$

Finally, check that $P + Q$ reproduces A exactly, confirming the decomposition is correct.

$$P + Q = \begin{bmatrix} 3 & 3 & 2.5 \\ 3 & 1 & 4.5 \\ 2.5 & 4.5 & 7 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 2.5 \\ 1 & 0 & -1.5 \\ -2.5 & 1.5 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 5 \\ 4 & 1 & 3 \\ 0 & 6 & 7 \end{bmatrix} = A \checkmark$$

Final Answer

$$A = \begin{bmatrix} 3 & 3 & 2.5 \\ 3 & 1 & 4.5 \\ 2.5 & 4.5 & 7 \end{bmatrix} + \begin{bmatrix} 0 & -1 & 2.5 \\ 1 & 0 & -1.5 \\ -2.5 & 1.5 & 0 \end{bmatrix} \text{ (symmetric + skew-symmetric)}$$

Common Mistake: Forgetting the $\frac{1}{2}$ factor and using $A + A'$ and $A - A'$ directly as the two parts – without halving, they don't add back to A at all, only to $2A$.

Quick Recall: Half of (A plus its transpose) is always symmetric; half of (A minus its transpose) is always skew-symmetric – and the two halves always add back to A .